Staying in Touch, Simply

Co-Designing a Way for an Older Adult to Use His Smartphone (mid-term draft)

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Many older adults own a smartphone they cannot use. We are co-designing with Manuel, an 80-year-old man who lives alone in Lisbon, a way for him to use the smartphone his daughter bought him so he could follow the family's WhatsApp group. He does not want a simpler phone: he wants the smartphone's family features, photos, messages and news, without having to learn how to work it. This mid-term draft covers related work, secondary research on older adults in Portugal, our method, and the first two of three co-design sessions, in which Manuel framed the problem and chose a concept: a plain cover that shows one thing at a time, with a green button to talk and a red one to end.

1 INTRODUCTION

Smartphones are now where many families share news and photos, but many older adults find them hard to learn and easy to get wrong [13, 14]. In Portugal, which has the second-oldest population in the EU, only one in five people aged 65–74 has at least basic digital skills [9, 10].

Rather than designing for older adults, we design with one of them [25]. Over three sessions following the Double Diamond [5], Manuel defines the problem with us, creates and chooses ideas, and will test our prototype at home. This draft covers the first two sessions.

2 RELATED WORK

2.1 Older adults and smartphones

Older adults are not simply afraid of technology: they weigh its benefits against its hassle [17], and anxiety, confidence and cognitive abilities, more than age itself, predict whether they adopt it [3]. Touchscreens remain a barrier. Older adults' typing errors on touchscreens rise with hand tremor [19], and flat designs make it harder for them to tell what can be pressed [28]. Senior launchers make the screen simpler, but fewer features alone do not make a phone usable [20].

2.2 Voice interaction

Voice assistants can reach older adults who avoid screens, but they are abandoned when people find few valuable uses [27], reminders go unused when people doubt them [23], and recognition errors become the main frustration over time [12]. A voice interface should do a few things the person values and recover gracefully from mistakes.

2.3 Co-design and stigma

Co-design treats the people who will use a design as co-creators [25], generative methods such as making workshops draw out people's own desires for technology [1], and participatory design helps older people take part from the start [16]. Designs must also avoid stigma: devices that mark their users as old or disabled are often rejected [18, 26].

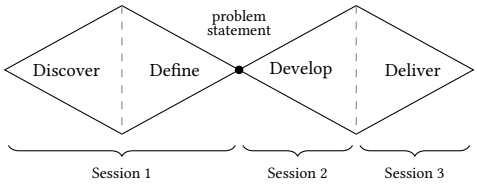


Fig. 1. Our three co-design sessions mapped onto the Double Diamond.

Table 1. Older adults in Portugal and the EU-27, Eurostat 2025 [6–10].

Indicator	Portugal	EU-27
Population aged 65 or over	24.3%	22.0%
Never used the internet, ages 65–74	36.1%	16.4%
At least basic digital skills, ages 65–74	20.7%	33.0%
Instant messaging, internet users aged 65–74	79.8%	74.5%
People aged 65 or over living alone	25.7%	32.2%

3 METHODOLOGY

3.1 Co-designer

Our co-designer Manuel is 80, a widower and a retired electrician, and lives alone in Lisbon; ~~his daughter introduced him to us. He calls his family every day on a button phone.~~ In July, his daughter bought him a smartphone so he could join the family's WhatsApp group, where photos of his newborn great-grandson are shared. He does not know how to work it, so it stays in a drawer.

3.2 Sessions

We follow the Double Diamond (Figure 1). Each session takes about 70 minutes in Manuel's kitchen. In session 1 we used directed storytelling, a tour of both his phones and a communication map, and wrote the problem statement with him. In session 2 he sorted 16 action cards, discussed possible ways to simplify his phone, acted out the phone's behaviour while a teammate played its voice in a Wizard of Oz setup [4], and narrated a storyboard. Session 3 will test our prototype.

We adapted the sessions to him: he hosted a card game before any question, we read the consent form aloud in plain Portuguese, and only one of us asked at a time. Participation follows a project declaration signed by Manuel and the team.

4 SECONDARY RESEARCH

Older adults in Portugal are less connected than their European peers (Table 1), but those who are online message and call more than the EU average, so staying in touch is a central use.

Existing products do not fit Manuel. Minimal phones such as the Light Phone drop apps like WhatsApp [15]. Simplified modes such as Samsung's Easy mode only change how the phone looks, and Apple's Assistive Access leaves apps like WhatsApp in their usual design [2, 24]. Amazon's and Google's smart speakers do not

Table 2. Manuel's card sort in session 2. * Cards he wrote himself.

Need	Would like	Don't need
• Only people my daughter saved*	• Appointment reminders	• Medication reminders
• See family photos	• Emergency help, only to call his daughter	• Torch
• Hear messages read aloud	• Video call, only the baby, his face hidden	• Radio: his late wife's station stays in the kitchen
• Know if a message is a scam	• Time and date	• Take photos
• Call someone, if as easy as the Nokia		• Weather
• Answer calls, same condition		• Magnifier
• Send a message by speaking, short phrases only		

support European Portuguese [11, 21], and Siri added it only in late 2025 [22].

5 EARLY FINDINGS

5.1 Session 1

- *He cannot work the smartphone.* His taps were too short, a pinch shrank the text and the keyboard opened by accident: "I broke the letters. Put them back."
- *But he wants what is inside it.* He will not refuse it, as his sister did, "because the refusal includes the baby".
- *He wants the name first.* After an "olá pai" scam text: "I still want to know the name before I see the child."

Quotes are our translations from Portuguese. Manuel corrected two drafts before signing this problem statement:

> Manuel needs a way to see the new photos of his great-grandson, to hear his great-granddaughter and the family group, and to reply with a short sentence, without learning the smartphone. He needs the sender's name first, because a message that says "olá pai" may be a scam.

5.2 Session 2

Manuel kept the cards about his family (Table 2) and wrote two of his own: the sender's name before any photo, and only people his daughter saved. We discussed different designs without showing him a physical prototype. He preferred a cover for the phone he already owned, allowing him to see one thing at a time without the usual icons. He asked for a green button to talk where his thumb rests and a red button to end.

In the role-play he set rules for the voice: no greeting, the sender's name first, silence while he looks, no photo from unknown senders and, after two misunderstandings, nothing at all: "Rude and safe. I prefer that to cheerful and wrong."

5.3 Lessons about the methods

Letting Manuel host a game before any question put him in charge, and options he could refuse made his choices clear. We were wrong that one button would be enough.

6 NEXT STEPS

Before session 3, we will develop a functional prototype for Manuel's smartphone, featuring a simplified interface with one item displayed at a time and clear green and red controls. The system will allow him to view family photos, hear messages read aloud in European Portuguese, and respond using short voice commands.

Unlike the role-play in session 2, Manuel will interact directly with a working system rather than simulated responses. This will allow us to evaluate its usability, identify difficulties during real interactions, and gather feedback for further improvements. Our goal is to determine whether the system can make accessing family communications as straightforward as using his existing button phone.

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